Notes: Campello, Graham, and Harvey (JFE, 2010)

The Real Effects of Financial Constraints: Evidence from a Financial Crisis

Contents

1 Big picture

- **Question.** The paper asks whether financial constraints during the 2008 financial crisis had real effects on firm behavior: investment, employment, technology spending, marketing, cash management, credit-line use, and asset sales.
- **Setting.** The authors survey 1,050 CFOs in the U.S., Europe, and Asia in late 2008, when credit markets were unusually tight.
- **Core empirical advantage.** Instead of using only archival proxies such as size, dividend status, or credit ratings, the paper directly asks managers whether the cost or availability of credit affected their firms.
- **Main finding.** Firms classified as financially constrained planned much deeper cuts in real activity than unconstrained firms. In the U.S., constrained firms planned to cut technology spending by 22%, marketing by 32.4%, employment by 10.9%, capital spending by 9.1%, cash holdings by 15.0%, and dividends by 14.2%.
- **Liquidity management.** Constrained firms burned through cash, drew more heavily on credit lines, and drew on credit lines for precautionary and strategic reasons, especially because they feared that banks might restrict future access.
- **Investment channel.** Credit constraints affected not only planned spending levels but also project choice. Many constrained CFOs said they bypassed attractive investment opportunities because external finance was unavailable.
- **Asset-sale channel.** Constrained firms were much more likely to sell assets to fund operations during the crisis.
- **Economic takeaway.** The paper provides direct survey evidence that the financial crisis changed real corporate decisions. The effects are strongest for firms that personally experienced credit-market frictions.

2 Incremental contribution of the paper

- **A direct measure of financial constraint.** The paper contributes by asking CFOs directly whether their firms were affected by the cost or availability of credit. This differs from the usual approach of inferring constraints from size, ratings, payout, or other balance-sheet variables.
- **Real-time crisis evidence.** The survey was conducted during the financial crisis, before many effects were fully visible in annual financial statements. This gives the paper near-real-time evidence on managerial plans and actions.
- **Broader outcome set.** The paper studies employment, technology spending, marketing spending, cash holdings, dividends, lines of credit, asset sales, and investment cancellation, not only capital expenditures.
- **International comparison.** The survey covers the U.S., Europe, and Asia, so the results show that the real effects of credit constraints were not limited to U.S. public firms.
- **Matching-estimator contribution.** The authors use matching estimators to compare constrained firms with otherwise similar unconstrained firms based on size, ownership, credit rating, profitability, dividend status, growth prospects, and industry.

- **Comparison with standard proxies.** A key contribution is showing that the direct survey measure identifies meaningful behavior during the crisis better than traditional proxies such as size, ownership form, and ratings.
- **Liquidity-management contribution.** The paper documents how firms actively managed liquidity in the crisis, especially through cash depletion and credit-line drawdowns.
- **Policy contribution.** The findings suggest that credit-market disruptions can reduce valuable investment and employment. Policies that unfreeze credit markets may therefore affect real activity, not only financial balance sheets.

3 Data and measurement

3.1 Survey design and sample

- The survey was administered in late 2008, near the peak of the credit crisis.
- The final sample includes 1,050 nonfinancial firms: 574 in the U.S., 192 in Europe, and 284 in Asia.
- The respondents are mainly CFOs, but the paper also groups treasurers, assistant treasurers, vice presidents of finance, comptrollers, and similar executives under the CFO label.
- The survey asks about planned changes over the next 12 months relative to the previous 12 months.
- Main planned-policy variables include technology expenditures, capital expenditures, marketing expenditures, employment, cash holdings, and dividend payments.

Interpretation: The survey design is useful because it captures managerial plans and constraints at the moment credit conditions tightened. The limitation is that it relies on self-reported answers and one survey wave for the most detailed crisis questions.

3.2 Survey-based financial constraint measure

- Firms are asked whether their operations were not affected, somewhat affected, or very affected by difficulties in accessing credit markets.
- The main binary classification treats the *VeryAffected* firms as financially constrained and the *NotAffected* and *SomewhatAffected* firms as unconstrained.
- In the U.S. survey, 244 firms report being not affected, 210 somewhat affected, and 115 very affected.
- The authors also ask constrained firms which type of friction they face: quantity constraints, higher borrowing costs, or difficulty initiating or renewing a credit line.
- Among *VeryAffected* firms, 81% report quantity constraints, 59% report higher costs, and 55% report difficult access to credit lines.

Interpretation: The direct constraint measure is not just a subjective label. It is tied to concrete experiences with credit rationing, higher prices of credit, and bank-line access problems.

3.3 Firm characteristics and comparability

- The paper compares survey firms with Compustat public firms along size, ratings, profitability, dividend payment, and cash holdings.
- The survey contains many private firms and non-U.S. firms, which broadens the sample beyond standard Compustat studies.
- For public U.S. survey respondents, cash-to-assets ratios are close to Compustat averages.
- Survey respondents are more likely to have credit ratings and investment-grade ratings than the full Compustat universe.

Interpretation: The sample is broadly informative but not perfectly representative. If anything, the public firms in the survey may be of somewhat higher quality than the typical Compustat public firm.

3.4 Policy and outcome measures

- The paper measures expected percentage changes in technology spending, capital spending, marketing, employment, cash holdings, and dividend payouts.
- It measures liquidity management using the level and change in cash balances and lines of credit relative to assets.
- It measures investment consequences by asking whether firms bypass attractive projects because external capital is limited.
- It measures financing choices when external finance is limited: cash flow, cash holdings, other sources, or canceling/postponing investment.
- It measures asset sales by asking whether firms are selling more assets than in previous years to obtain funding.

Interpretation: The outcomes cover both planned real activity and realized financial responses. This lets the paper trace a credit-supply shock into multiple corporate margins.

4 Models and empirical specifications

4.1 Mean comparisons by firm groups

The paper first reports simple differences in planned policies across groups:

$$\Delta Policy_i = \text{planned percentage change for firm } i.$$

- The authors compare means across geography, size, public versus private status, rating category, and survey-based constraint status.
- These comparisons provide the raw pattern: constrained firms plan larger cuts across nearly all margins.
- The figures are descriptive and do not fully control for other differences across firms.

Interpretation: The raw comparisons are useful for establishing the main crisis pattern, but they do not by themselves solve selection concerns.

4.2 Matching estimator for survey-based constraints

The core matching exercise compares constrained firms to otherwise similar unconstrained firms. Conceptually, the estimand is an average treatment effect for treated firms:

$$ATT = E[Y_i(1) - Y_i(0) \mid Constrained_i = 1].$$

- The treatment is being financially constrained according to the CFO survey response.
- In the pre-crisis and crisis comparison, firms are matched on size, ownership form, credit rating, industry, and survey quarter.
- In the crisis-only finer matching, firms are additionally matched on profitability, dividend payer status, and long-term growth prospects.
- The paper reports both the Abadie-Imbens matching estimator and the Dehejia-Wahba propensity-score estimator.

Interpretation: Matching tries to compare firms that look similar on observed characteristics but differ in their direct credit-constraint status. This makes it harder to attribute the results only to size, rating, ownership, or industry composition.

4.3 Comparison with standard proxies

The paper repeats the matching exercise using traditional proxy definitions of constraints:

$$Small - Large, \qquad Private - Public, \qquad Speculative\text{-}grade - Investment\text{-}grade.$$

- These proxy-based treatments are then compared with the survey-based treatment.
- The traditional proxies often have weak, noisy, or counterintuitive relations with corporate plans during the crisis.
- The survey measure is more consistent in identifying firms that cut spending and liquidity more sharply.

Interpretation: The comparison is important because it validates the survey measure. The direct measure captures crisis-specific financing frictions that standard archival proxies miss.

4.4 Liquidity-management tests

- Cash holdings are compared one year before the crisis and at the crisis peak.
- Credit-line availability is also compared one year before the crisis and at the crisis peak.
- The paper asks firms why they draw down credit lines: liquidity needs, daily operations, precautionary motives, and strategic timing.

Interpretation: The liquidity tests show that constrained firms did not merely report lower investment plans. They actively used cash and bank lines to survive the crisis.

4.5 Investment and asset-sale tests

- The investment tests ask whether access to external capital limited the ability to pursue attractive investment projects.
- The paper compares the responses in normal times and during the crisis period.
- The asset-sale tests ask whether firms are selling more assets than in previous years to obtain funding.

Interpretation: These tests translate financial frictions into real decisions: project cancellation, reliance on internal funds, and sale of assets.

5 Identification and empirical design

5.1 What the design can identify

- The design identifies associations between self-reported financing constraints and corporate policies during a severe aggregate credit shock.
- The matching estimators improve comparability but do not convert the survey into a randomized experiment.
- The evidence is strongest for showing that firms reporting concrete credit frictions also report systematically different real and financial actions.

Interpretation: The paper should be read as strong crisis-period evidence that credit constraints matter for firm behavior, not as a perfectly randomized causal experiment.

5.2 Why the crisis setting helps

- The 2008 crisis created a large negative shock to credit supply.
- The shock makes financing constraints salient and observable to managers.
- It also creates meaningful cross-sectional variation because some firms could still access credit while others could not.

Interpretation: The crisis is a useful empirical laboratory because it reveals which firms are vulnerable to credit-market disruptions.

5.3 Key concerns

- **Subjective reporting.** CFOs may perceive constraints differently, and some responses may reflect pessimism or demand concerns.
- **Firm heterogeneity.** Poor-performing firms may both invest less and have worse access to credit.
- **Representativeness.** The survey sample is broad but not a random sample of all firms.

- **One-time survey limits.** The most detailed liquidity and investment questions are measured at one point during the crisis.

Interpretation: The authors address these concerns with matching and with evidence that constrained firms report tangible credit-market problems. Still, the design cannot rule out all unobserved heterogeneity.

6 Tables and figures

6.1 Table 1: Survey invitations and U.S. response rates

Read:

- Table 1 reports the distribution of U.S. survey invitations and responses by annual sales and industry.
- Response rates are roughly in the 5% to 8% range across major size and industry categories, with a higher response rate in the "Other" category.
- The table shows that the survey was not confined to one narrow industry or one firm-size class.

Economic message: The survey has broad corporate coverage, which is essential because the paper studies whether financing frictions affect many types of firms during the crisis.

6.2 Table 2: Comparing public survey companies to Compustat

Read:

- Table 2 compares U.S. public survey firms with nonfinancial public Compustat firms.
- Survey firms are somewhat more likely to be large, rated, and investment grade.
- Profitability, dividend-paying status, and cash-to-assets ratios are fairly comparable across the two samples.

Economic message: The survey sample is not identical to the Compustat universe, but it is close enough on several key dimensions to make the results informative for standard corporate-finance questions.

Table 1

Survey invitations and U.S. response rates.

This table reports the firm size and industry breakdowns of the 10,000 survey E-mail invitations sent to U.S. firms by *CFO* magazine in 2008Q4. The table also reports the number of respondents and the response rates. The reported numbers reflect "bounce backs" (nearly 7%) and include financial firms (which are excluded from the main analysis).

Characteristic	Category	Survey invitations (N)	Surveys received (N)	Response rate (%)
Annual sales volume	< $1 Billion	6,813	509	7.5%
	> $1 Billion	3,187	170	5.3%
Industry	Retail/wholesale	1,112	87	7.8%
	Manufacturing	2,321	144	6.2%
	Transportation/energy	573	42	7.3%
	Communications/media	372	26	7.0%
	Technology	521	24	4.6%
	Banking/finance/insurance	2,308	105	4.5%
	Service/consulting	691	45	6.5%
	Healthcare/pharmaceutical	743	51	6.9%
	Other	1,226	141	11.5%

(a) Table 1: Survey invitations and U.S. response rates

Table 2

Comparing public survey companies to Compustat.

This table contrasts observations (raw counts) and frequencies (in percentage terms) of firms in the U.S. survey sample and those in Compustat as of 2008Q4. The samples are restricted to non-financial public companies. Firms are considered to be "large" if their annual sales surpass $1 billion, and "small" otherwise. "Speculative" firms are those with S&P credit ratings equal to BB+ or below. "Investment-grade" firms have ratings of BBB− or above. Dividends and profits refer to fiscal year 2008. Cash/Assets is the ratio of cash and liquid securities to total assets.

Observable	Category	Survey sample Obs. (N)/Freq. (%)	Compusat sample Obs. (N)/Freq. (%)
Size	Small	73/56%	3,436/69%
	Large	57/44%	1,543/31%
Credit rating	Speculative	26/27%	698/52%
	Investment	70/73%	635/48%
Profitability	Profits > 0	110/87%	3,961/80%
	Profits ≤ 0	16/13%	1,018/20%
Dividend payments	Dividends > 0	59/47%	1,977/40%
	Dividends = 0	67/53%	3,002/60%
		Mean/median	Mean/median
Cash/Assets		0.163/0.080	0.170/0.083

(b) Table 2: Survey firms versus Compustat firms

6.3 Figure 1: Planned policies by geography and firm characteristics

Read:

- Figure 1 shows planned changes in technology expenditures, capital expenditures, marketing, employment, cash holdings, and dividends.
- Across U.S., Europe, and Asia, firms planned cuts in most categories.
- In the U.S., speculative-grade firms planned larger reductions than investment-grade firms.
- Public firms planned larger technology-spending cuts, while private firms planned larger capital-spending and marketing cuts.

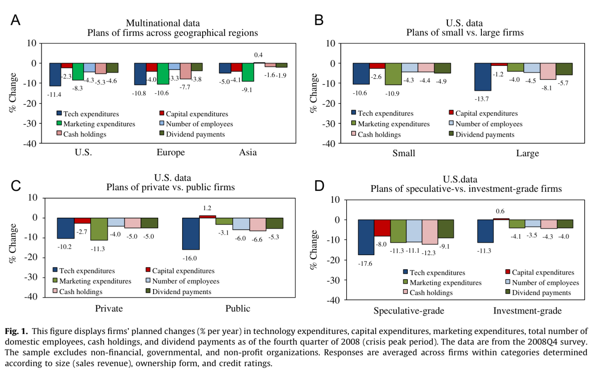
Economic message: The financial crisis affected many corporate policies at once. Rating-based differences already suggest that access to credit matters, but demographics alone do not fully capture the constraint channel.

6.4 Figure 2: Planned policies by survey-based constraint status

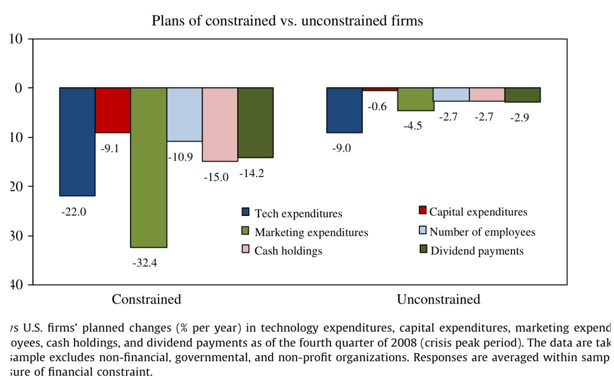
Read:

- Figure 2 compares constrained and unconstrained U.S. firms using the direct survey measure.
- Constrained firms planned much larger cuts in technology, capital spending, marketing, employment, cash, and dividends.
- The difference is especially dramatic for marketing spending and technology spending.

Economic message: The survey-based constraint measure sharply separates firms that cut real activity from firms that did not. This is the paper's central descriptive result.



(a) Figure 1: Planned policies by geography, size, ownership, and rating



(b) Figure 2: Planned policies by survey-based constraint status

6.5 Table 3: Sample descriptives across constraint types

Read:

- Table 3 describes firms that are *NotAffected*, *SomewhatAffected*, and *VeryAffected* by credit constraints.

- Size and ownership are only weakly related to the survey constraint measure.
- Credit ratings, profitability, and growth prospects are more related to constraint status, but the relation is far from perfect.

Economic message: Standard observables do not fully explain which firms report being constrained. This motivates using the direct survey measure.

6.6 Table 4: Characterizing financing frictions

Read:

- Table 4 asks constrained firms what type of credit friction they face.
- Among *VeryAffected* firms, 81% report quantity constraints, 59% report higher costs, and 55% report difficult line-of-credit access.
- Frictions are less severe for *SomewhatAffected* firms.

Economic message: The survey constraint measure corresponds to concrete credit-market obstacles. The constrained group is not just more pessimistic; it reports actual difficulty obtaining external finance.

Table 3

Sample descriptives across constraint types.

This table reports the number of observations (raw counts) and frequencies (in percentage terms) of relevant characteristics of firms that declare themselves as *NotAffected*, *SomewhatAffected*, and *VeryAffected* by credit constraints and by firm types. Firms are considered to be "large" if their annual sales surpass $1 billion, and "small" otherwise. "Speculative" firms are those with S&P credit ratings equal to BB+ or below. "Investment" firms have ratings of BBB− or above. Dividends and profits refer to fiscal year 2008. Growth prospects reflect CFOs' views about the long-term growth prospects of their firms on a one-to-ten scale. The data are taken from the 2008Q4 U.S. survey and exclude non-financial, governmental, and non-profit organizations. Column 1, for example, shows that 179 (or 73%) of the firms in the *NotAffected* category are small, while 65 (27%) are large. For the *SomewhatAffected* category, 78% of the firms are small and 22% are large, while for the *VeryAffected* category the breakdown is 82% small and 18% large. The numbers in row 1 also describe the proportion of constraint-types "within" the size category. For the small category (436 firms), 41% of the firms are *NotAffected*, 37% are *SomewhatAffected*, and 22% are *VeryAffected*. For the large firms (133 firms), 49% of the observations are *NotAffected*, 35% are *SomewhatAffected* and 16% are *VeryAffected*.

Observable	Category	*NotAffected* N/% across/% within	*SomewhatAffected* N/% across/% within	*VeryAffected* N/% across/% within
Size	Small	179/73%/41%	163/78%/37%	94/82%/22%
	Large	65/27%/49%	47/22%/35%	21/18%/16%
Ownership	Private	142/70%/42%	121/73%/36%	74/76%/22%
	Public	61/30%/47%	45/27%/35%	24/24%/18%
Credit rating	Speculative	6/15%/23%	8/25%/31%	12/57%/46%
	Investment	35/85%/52%	24/75%/35%	9/43%/13%
Profitability	Profits > 0	208/90%/47%	156/80%/35%	82/71%/18%
	Profits ≤ 0	24/10%/25%	40/20%/41%	33/29%/34%
Dividend pay	Dividends > 0	76/36%/46%	60/35%/36%	30/30%/18%
	Dividends = 0	133/64%/43%	111/65%/35%	70/70%/22%
Growth prospects	Prospects > 5	193/79%/45%	161/77%/37%	77/67%/18%
	Prospects ≤ 5	50/21%/36%	49/23%/36%	38/33%/28%

(a) Table 3: Constraint status and firm characteristics

Table 4

Characterizing financing frictions across constraint types.

This table reports the number of observations (raw counts) and frequencies (in percentage terms) of different forms of credit frictions faced by firms that declare themselves as *SomewhatAffected* and *VeryAffected* by credit constraints. Quantity constraint indicates whether the firm has experienced less access to credit. Price constraint indicates whether the firm has experienced higher cost of funds. Difficult access to LC indicates whether the firm has found difficulties in obtaining new lines of credit or renewing existing ones. The data are from the 2008Q4 U.S. survey.

Observable	Category	*SomewhatAffected* Obs. (N)/Freq. (%)	*VeryAffected* Obs. (N)/Freq. (%)
Quantity constraint	No	105/50%	22/19%
	Yes	105/50%	93/81%
Price constraint	No	125/60%	47/41%
	Yes	85/40%	68/59%
Difficult access to LC	No	169/80%	52/45%
	Yes	41/20%	63/55%

(b) Table 4: Types of financing frictions

6.7 Table 5: Matching estimates for the direct constraint measure

Read:

- Table 5 reports average treatment effects from matching estimators.
- Constrained firms planned deeper cuts even before the crisis, but the differences become much larger in 2008Q4.
- During the crisis, constrained firms planned around 8 to 12 percentage points more cuts in capital, technology, marketing, and employment, and much larger dividend cuts.

Economic message: The crisis widened the gap between constrained and unconstrained firms. Financial constraints matter more when external credit markets are stressed.

6.8 Table 6: Finer matching during the crisis

Read:

- Table 6 adds controls for profitability, dividend payout, growth prospects, and industry.
- The constrained-unconstrained differences remain economically large and statistically meaningful.
- Constrained firms still plan substantially larger cuts in technology, capital expenditures, marketing, employment, cash, and dividends.

Economic message: The main results are not simply driven by constrained firms having worse profitability or weaker growth opportunities.

Table 5

Corporate policies: average treatment effects (matching estimators) for the direct measure of financial constraint over pre-crisis and crisis periods.

This table reports differences in planned annual percentage changes of real and financial corporate policies according to whether firms are financially constrained or financially unconstrained. The financial constraint measure is based on self-reported difficulty in accessing credit. Differences are computed as average treatment effects via matching estimators (ATT). Firms are matched across the demographics of asset size, ownership form, and credit ratings. Columns 1 and 2 report results for the pre-crisis period (2007Q3 through 2008Q3). Columns 3 and 4 report results for the crisis period (2008Q4). The data are from the U.S. surveys. The Abadie and Imbens (2002) estimates are obtained from the bias-corrected, heteroskedasticity-consistent estimator implemented in Abadie et al. (2004). The Dehejia and Wahba (2002) estimates are obtained from the nearest neighbor matching estimator implemented in Becker and Ichino (2002), imposing the common support condition and using bootstrapped errors. *t*-Statistics are in parentheses.

Policy	Difference between constrained and unconstrained firms			
	Pre-crisis period		Crisis period	
	Abadie-Imbens	Dehejia-Wahba	Abadie-Imbens	Dehejia-Wahba
% Change in technology expenditures	−5.467*** (−2.61)	−5.369*** (−2.72)	−11.160*** (−3.09)	−11.278*** (−3.00)
% Change in capital expenditures	−7.706*** (−2.57)	−7.813*** (−2.63)	−8.494*** (−3.79)	−8.054*** (−2.73)
% Change in marketing expenditures	−5.878*** (−3.19)	−5.843*** (−3.19)	−11.709*** (−4.05)	−11.866*** (−3.75)
% Change in employees	−5.603*** (−4.04)	−5.541*** (−3.43)	−8.431*** (−4.18)	−8.495*** (−3.89)
% Change in cash holdings	−3.467 (−1.39)	−3.589 (−1.58)	−8.536* (−1.87)	−8.496** (−2.03)
% Change in dividend payout	−7.559** (−1.98)	−7.172* (−1.70)	−28.412** (−2.09)	−27.941** (−1.97)

Note: ***, **, and * indicate statistical significance at the 1%, 5%, and 10% (two-tail) test levels.

(a) Table 5: Matching estimates for the direct constraint measure

Table 6

Firm heterogeneity and the impact of financial constraints on firm plans during the financial crisis.

This table reports differences in planned annual percentage changes of real and financial policies of firms according to whether they are financially constrained or financially unconstrained. The financial constraint measure is based on self-reported difficulty in accessing credit. Differences are computed as average treatment effects via matching estimators (ATT). Firms are matched across demographic, financial, and economic characteristics: asset size, ownership form, credit ratings, profitability, dividend payout, long-term investment prospects, and industry. The data are collected from the 2008Q4 U.S. survey. The Abadie and Imbens (2002) estimates are obtained from the bias-corrected, heteroskedasticity-consistent estimator implemented in Abadie et al. (2004). The Dehejia and Wahba (2002) estimates are obtained from the nearest neighbor matching estimator implemented in Becker and Ichino (2002), imposing the common support condition and using bootstrapped errors (500 repetitions). *t*-Statistics are in parentheses.

Policy	Diff. between constrained and unconstrained firms Crisis period	
	Abadie-Imbens	Dehejia-Wahba
% Change in technology expenditures	−11.468*** (−2.69)	−12.955*** (−2.89)
% Change in capital expenditures	−7.581*** (−2.59)	−6.822** (−2.21)
% Change in marketing expenditures	−12.424*** (−4.15)	−13.240*** (−3.88)
% Change in employees	−5.977*** (−3.90)	−5.326*** (−2.65)
% Change in cash holdings	−7.666* (−1.69)	−9.006** (−2.07)
% Change in dividend payout	−28.640** (−2.28)	−28.392** (−1.99)

Note: ***, **, and * indicate statistical significance at the 1%, 5%, and 10% (two-tail) test levels.

(b) Table 6: Finer matching during the crisis

6.9 Table 7: Traditional measures of financial constraints

Read:

- Table 7 repeats matching tests using size, ownership form, and credit ratings as traditional constraint proxies.
- The estimates are much less consistent than the survey-based constraint measure.
- Some signs are counterintuitive during the crisis, such as small or speculative-grade firms sometimes appearing to cut less than comparison groups.

Economic message: Traditional proxies are imperfect in this crisis setting. They do not reliably capture which firms are actually facing binding financing frictions.

6.10 Table 8: Reasons firms draw down lines of credit

Read:

- Table 8 reports why firms draw down lines of credit.
- Constrained firms are more likely to draw for liquidity needs and daily operations.
- They are also more likely to draw for precautionary reasons and strategic timing, including fear that banks will limit future access.

Economic message: Credit lines become a crisis-management tool. Constrained firms use them not only for routine liquidity but also to protect against future credit rationing.

Table 7

Corporate policies: average treatment effects (matching estimators) for traditional measures of financial constraints over pre-crisis and crisis periods.

This table reports differences in planned annual percentage changes of real and financial corporate policies according to whether firms are financially constrained or financially unconstrained. Three traditional financial constraint measures are considered: firm size (small minus large), ownership form (private minus public), and credit ratings (speculative minus investment grade). Differences are computed as average treatment effects via matching estimators (ATT). Columns 1 through 3 report results for the pre-crisis period (2007Q3 through 2008Q3). Columns 4 through 6 report results for the crisis period (2008Q4). The data are from the U.S. surveys. The table uses Abadie and Imbens (2002) estimates that are obtained from the bias-corrected, heteroskedasticity-consistent estimator implemented in Abadie et al. (2004). *t*-Statistics are in parentheses.

Policy	Difference between "constrained" and "unconstrained" firms					
	Pre-crisis period			Crisis period		
	Size	Ownership	Ratings	Size	Ownership	Ratings
% Change in technology expenditures	2.304 (1.21)	−1.547 (−1.03)	−4.877** (−2.04)	5.775 (0.87)	0.028 (0.01)	12.601 (1.10)
% Change in capital expenditures	3.646 (1.24)	−2.034 (−0.79)	−7.621** (−2.24)	2.246 (0.24)	8.902* (1.80)	15.903 (1.26)
% Change in marketing expenditures	2.528* (1.92)	−0.034 (−0.03)	−2.980 (−1.24)	15.259 (0.91)	−7.873 (−0.67)	−12.763 (−1.04)
% Change in employees	2.640*** (2.79)	0.426 (0.52)	1.723 (1.29)	−6.479 (−1.54)	2.074 (0.79)	−9.202* (−1.73)
% Change in cash holdings	4.885* (1.86)	−3.738 (−1.71)	−2.399 (−0.79)	2.372 (0.11)	−5.801 (−0.47)	24.826 (0.67)
% Change in dividend payout	−0.615 (−0.18)	0.022 (0.14)	−4.508 (−1.59)	28.022* (1.96)	−6.183 (−1.04)	−13.041 (−0.44)

Note: ***, **, and * indicate statistical significance at the 1%, 5%, and 0% (two-tail) test levels.

(a) Table 7: Traditional proxy measures of constraints

Table 8

Why do firms draw down their lines of credit?

This table displays standard mean comparison tests (OLS) for the proportion of constrained and unconstrained firms reporting each of the rationales listed as a reason for using their LCs. The strategic timing row shows the portion of respondents that draw down on their credit lines in December 2008 because they were afraid their banks would limit credit line access in the future. The data are collected from the 2008Q4 U.S. survey. *t*-Statistics are in parentheses.

Policy	Constrained	Unconstrained	Diff. const. − unconst.
Liquidity needs	0.504*** (10.77)	0.282*** (13.34)	0.222*** (4.62)
Daily operations	0.548*** (11.75)	0.421*** (18.14)	0.127** (2.46)
Precautionary	0.130*** (4.14)	0.068*** (5.76)	0.062** (2.19)
Strategic timing	0.165*** (4.75)	0.059*** (5.35)	0.106*** (3.76)

Note: ***, **, and * indicate statistical significance at the 1%, 5%, and 10% (two-tail) test levels.

(b) Table 8: Why firms draw down credit lines

6.11 Figure 3: Cash management during the crisis

Read:

- Figure 3 compares cash holdings one year before the crisis and at the crisis peak.
- Constrained U.S. firms reduce cash from 15.1% of assets to 11.7%.
- Unconstrained U.S. firms slightly increase cash from 14.6% to 14.8%.
- Similar cash-burn patterns appear in Europe and, to a lesser extent, Asia.

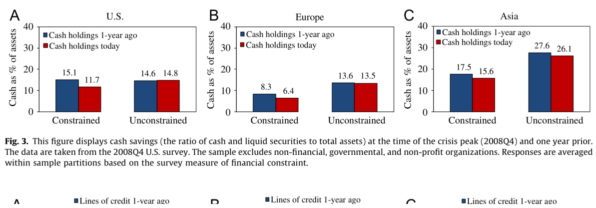
Economic message: Financially constrained firms use cash reserves as a buffer when external credit dries up. The crisis turns cash from precautionary savings into operating liquidity.

6.12 Figure 4: Lines of credit during the crisis

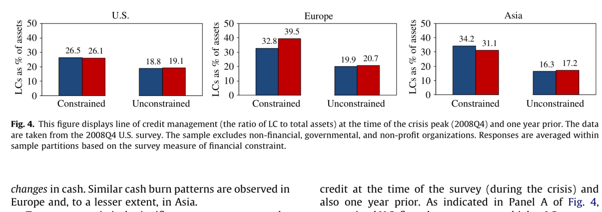
Read:

- Figure 4 compares lines of credit one year before the crisis and at the crisis peak.
- Constrained firms generally have higher line-of-credit-to-asset ratios than unconstrained firms.
- The most notable increase appears for constrained European firms, whose ratio rises from 32.8% to 39.5%.

Economic message: Lines of credit are an important source of liquidity during the crisis, but the drawdown evidence in Table 8 shows that firms worry about whether banks will continue to honor these facilities.



(a) Figure 3: Cash holdings before and during the crisis



(b) Figure 4: Lines of credit before and during the crisis

6.13 Table 9: External finance and investment

Read:

- Table 9 asks whether access to external capital limits firms' ability to pursue attractive investment projects.
- In normal times, only the credit-rating and financial-constraint categories show large differences.
- During the crisis, financially constrained firms show a much higher propensity to report that credit frictions limit attractive investment.
- The key direct-constraint difference rises to 0.419 in the crisis period.

Economic message: The crisis increases the real cost of financing constraints: firms pass up valuable projects because they cannot raise external funds.

6.14 Figure 5: How firms fund investment when external capital is limited

Read:

- Figure 5 shows how firms fund investment when external capital is limited.
- Firms use internal cash flow and cash holdings, but many also cancel or postpone investment.
- Constrained firms are more likely to cancel or postpone investment, especially in Europe and Asia.
- In the U.S., 56% of constrained firms report investment cancellation or postponement, compared with 31% of unconstrained firms.

Economic message: Credit constraints do not simply change the source of funds. They can eliminate investment altogether.

Table 9
The effects of credit frictions on corporate investment: "Normal times" and the "crisis period".

This table displays standard mean comparison tests (OLS) for the proportion of firms reporting that their ability to invest in attractive investment projects is limited by their access to the capital markets across different group categories. Category 1 groups firms that are small, private, speculative-grade rated, and financially constrained. Category 2 groups firms that are large, public, investment-grade rated, and financially unconstrained. The data are from the 2008Q4 U.S. survey. *t*-Statistics are in parentheses.

Criteria	Category 1	Category 2	Diff. categories
Panel A: Normal times			
By size	0.255***	0.236***	0.019
	(12.19)	(6.241)	(0.43)
By ownership	0.246***	0.272***	−0.026
	(10.47)	(6.81)	(−0.58)
By ratings	0.533***	0.190***	0.344***
	(5.76)	(5.19)	(4.00)
By fin. constraint	0.464***	0.200***	0.265***
	(9.81)	(10.53)	(5.93)
Panel B: Crisis period			
By size	0.514***	0.539***	−0.025
	(21.35)	(12.19)	(0.50)
By ownership	0.512***	0.544***	−0.032
	(18.74)	(12.16)	(0.61)
By ratings	0.800***	0.487***	0.313***
	(10.77)	(10.41)	(3.15)
By fin. constraint	0.857***	0.438***	0.419***
	(25.81)	(18.61)	(8.41)

Note: ***, **, and * indicate statistical significance at the 1%, 5%, and 10% (two-tail) test levels.

(a) Table 9: Credit frictions and investment

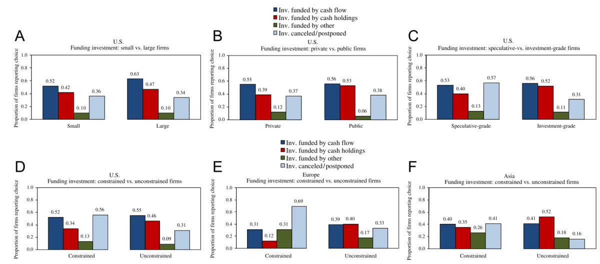


Fig. 5. This figure displays the proportion of firms indicating how they fund their investment when external capital is limited. The data are taken from the 2008Q4 U.S. survey. The sample excludes non-financial, governmental, and non-profit organizations. Responses are averaged within sample partitions based on size, ownership, credit ratings, and financial constraint.

(b) Figure 5: Funding or canceling investment when external capital is limited

6.15 Table 10: Asset sales during the crisis

Read:

- Table 10 asks whether firms sell assets to obtain funding during the crisis.
- Differences by size and ownership are near zero.
- The financial-constraint category is sharply different: 70% of constrained firms report more asset sales versus 36.6% of unconstrained firms.
- The difference by financial constraint is 0.334 and statistically significant.

Economic message: Financially constrained firms liquidate assets to fund operations. This creates a real asset-market channel through which credit-market stress can affect firms.

Table 10
Do firms sell assets to obtain funding during the crisis?

This table displays mean comparisons for the proportion of firms reporting that they are currently selling more assets than in previous years across group categories. Category 1 groups firms that are small, private, speculative-grade rated, and financially constrained. Category 2 groups firms that are large, public, investment-grade rated, and financially unconstrained. The data are collected from the 2008Q4 U.S. survey. *t*-Statistics are in parentheses.

Criteria	Category 1	Category 2	Diff. categories
By size	0.476*** (9.72)	0.478*** (6.42)	−0.002 (−0.02)
By ownership	0.489*** (9.23)	0.489*** (6.49)	−0.001 (−0.01)
By ratings	0.588*** (4.78)	0.415*** (5.32)	0.174 (1.20)
By fin. constraint	0.700*** (10.69)	0.366*** (7.60)	0.334*** (4.04)

Note: ***, **, and * indicate statistical significance at the 1%, 5%, and 10% (two-tail) test levels.

Table 10: Asset sales to obtain funding during the crisis

7 Short summary

- The paper surveys 1,050 CFOs during the 2008 financial crisis.
- It directly measures whether firms were financially constrained by asking managers about credit-market access.
- Constrained firms planned much larger cuts in investment, technology, marketing, employment, cash, and dividends.
- Matching estimators show that the results survive controls for firm size, ownership, ratings, profitability, dividend status, growth prospects, and industry.
- Traditional constraint proxies perform poorly relative to the direct survey measure.
- Constrained firms burned through cash and drew down credit lines more aggressively.
- Credit-line drawdowns reflect both liquidity needs and fears that banks would restrict future access.
- Constrained firms were more likely to bypass attractive investment, cancel or postpone projects, and sell assets.
- The paper's main conclusion is that financial constraints during the crisis had direct real effects on corporate behavior.

8 Conclusion

8.1 Core conclusion

Campello, Graham, and Harvey show that firms facing financing constraints during the 2008 financial crisis changed real and financial policies in economically important ways. Constrained firms planned deeper spending cuts, reduced employment more, used more cash, drew more heavily on credit lines, canceled investments, and sold assets to obtain funding.

8.2 Economic intuition

The economic intuition is straightforward. When external finance is available, firms can smooth temporary shocks and continue funding positive-NPV projects. When credit markets seize up, constrained firms must rely on internal funds and existing liquidity. If those internal resources are insufficient, they cut spending, reduce hiring, cancel projects, draw on lines of credit, or sell assets. The crisis turns financing frictions into real distortions.

8.3 Incremental contribution revisited

The paper advances the financial-constraints literature by replacing indirect classification with direct managerial evidence. Rather than assuming that small firms or unrated firms are constrained, the authors ask CFOs whether credit-market access affected operations. This approach makes it possible to compare the direct measure with standard proxies and to study policy margins that archival data rarely reveal.

8.4 Implications for corporate finance

The findings imply that liquidity management is central during crises. Cash reserves and credit lines act as buffers, but they are not perfect substitutes for functioning credit markets. Firms that cannot obtain credit may forgo attractive projects and liquidate assets, even when those decisions are costly.

8.5 Implications for policy

The results suggest that policies aimed at restoring credit supply can have real effects. If constrained firms cancel investment and reduce employment because they cannot finance attractive projects, then credit-market interventions can influence employment, investment, and growth.

8.6 Limitations

The analysis is based on survey responses, so measurement error and managerial perceptions may matter. Matching controls for many observable differences but cannot eliminate all unobservable heterogeneity. The sample is broad but not random, and some findings rely on one crisis-period survey wave.

8.7 Takeaway

The enduring takeaway is that financial constraints are not only accounting labels or balance-sheet correlations. During the 2008 crisis, firms that reported binding credit frictions made different real decisions. The paper connects credit-market disruptions to corporate investment, employment, liquidity, and asset-sale behavior in a direct and timely way.